Data Flow Diagram

Multimodal Attendance & Engagement Agent • Nirbhay Dubey | B.Tech CSE (Year 1)

The data flow diagrams show how camera frames and audio chunks become attendance records, engagement scores, and recommendations, and where data is stored. Everything runs on the local machine.

1. Level 0 - Context Diagram

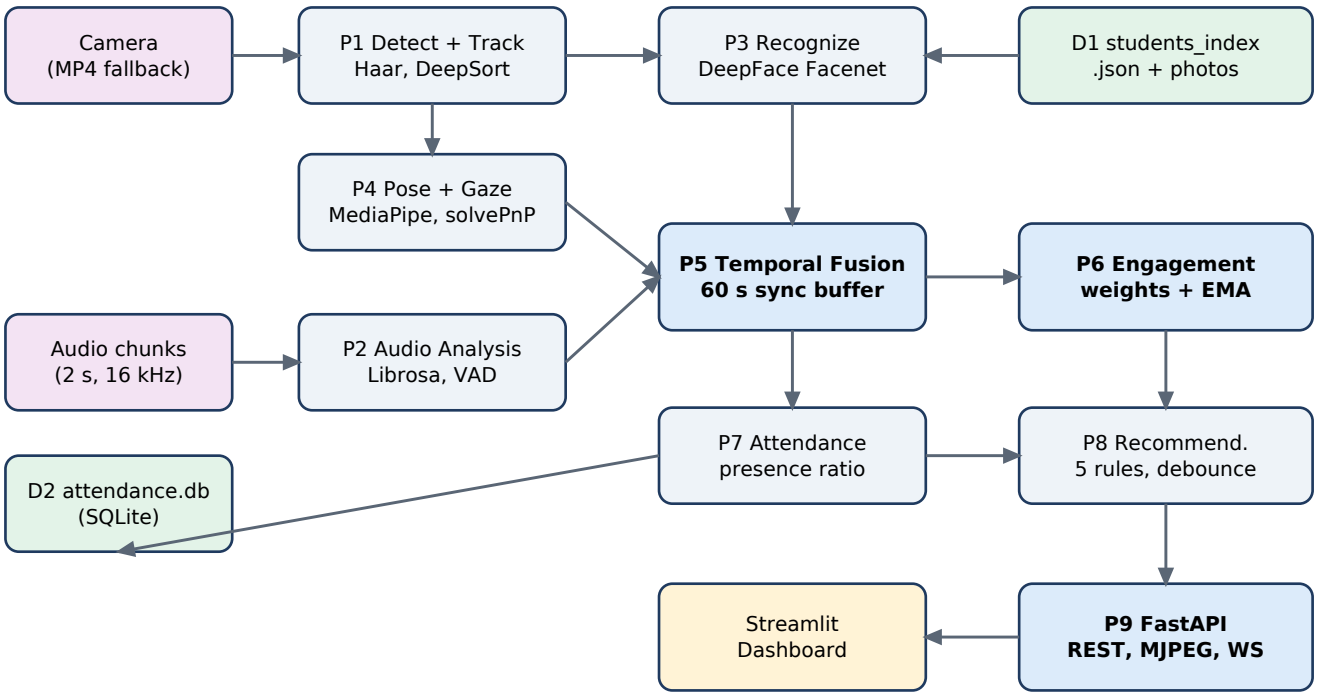
Level 0: System context



Students are observed through the camera and microphone; the lecturer interacts only with the dashboard.

2. Level 1 - Main Processes

Level 1: Processes and data stores



Data flow table

Flow	From	To	Data
F1	Camera	P1	BGR video frames (640x480)
F2	P1	P3 / P4	Face boxes, track IDs, face crops
F3	D1	P3	Enrolled 128-d Facenet embeddings
F4	Microphone	P2	2 s audio chunks posted to /api/audio/chunk
F5	P2	P5	Acoustic state, RMS, dBFS, SNR
F6	P3, P4	P5	Student ID or anonymous ID; yaw, pitch, roll, gaze, EAR
F7	P5	P6 / P7	Time-aligned multimodal observation per student
F8	P6, P7	P8	Class average engagement, forward-gaze ratio, attendance rate
F9	P7, P8	D2	Attendance records and recommendations (on session events)
F10	P9	Dashboard	JSON status, MJPEG video, WebSocket telemetry at 4 Hz

Per-frame metrics stay in memory during a session; the person_metrics table exists but is not written during live streaming.